LAB MANUAL

K.R. MANGALAM UNIVERSITY

SOET

(School of Engineering & Technology)

B.TECH CSE (CORE)

Database & Management System Lab (ENCS254)

Submitted by-

Akansha

Roll no - 2401010195

Submitted to-

Miss Mansi Kajal

Objective: To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

DDL QUESTIONS:

1. How do you create a table named CLASS with appropriate data types?
2. How do you create a table named EMPLOYEES with the specified attributes?
3. How do you create a table named Client_master with the specified attributes?

SOLUTION:

```sql
CREATE DATABASE LABSHEET2;
USE LABSHEET2;

CREATE TABLE CLASS(
ID INT PRIMARY KEY,
Name VARCHAR(20)
);
INSERT INTO CLASS VALUES (1,'Bravo');
INSERT INTO CLASS VALUES (2,'Alex');
INSERT INTO CLASS VALUES (4,'Cheng');
SELECT*FROM CLASS;
CREATE TABLE EMPLOYEES (
    Employee_Id INT,
    First_Name VARCHAR(20),
    Last_Name VARCHAR(20),
    Email VARCHAR(30),
    Phone_Number VARCHAR(15),
    Hire_Date DATE,
    Job_Id VARCHAR(15),
    Salary INT,
    Commission_Pct INT,
    Manager_Id INT,
    Department_Id INT
);
```

```sql
INSERT INTO EMPLOYEES VALUES
(101,'Akansha','Yadav','ay@gmail.com','1234567890','2023-08-16','HR',500000, NULL, 100, 15);
INSERT INTO EMPLOYEES VALUES
(102,'Dhruv','Kumar','dhrvk@gmail.com','1234567891','2023-05-21','IT',100000, NULL, 77, 21);
INSERT INTO EMPLOYEES VALUES
(103,'Kishan','Verma','kv@gmail.com','1234567892','2021-1-09','Law',900000, NULL, 1, 16);
INSERT INTO EMPLOYEES VALUES
(104,'Neha','Yadav','nehay@gmail.com','1234567893','2022-11-11','Sales',850000, NULL, 90, 25);
INSERT INTO EMPLOYEES VALUES
(105,'Simon','Jangra','simon@gmail.com','1234567894','2023-9-09','Finance',220000, NULL, 19, 7);
SELECT*FROM EMPLOYEES;
CREATE TABLE Client_master (
    ClientNo VARCHAR(10),
    Name VARCHAR(20),
    Address VARCHAR(30),
    City VARCHAR(20),
    State VARCHAR(20),
    Pincode INT,
    Bal_due INT
);
```

4. What SQL statement would you use to rename the table Client_master to Client12?

SOLUTION:

```sql
RENAME TABLE Client_Master TO Client12;
```

5. How do you display the bal_due column heading as “BALANCE” ?

```sql
SELECT Bal_due AS BALANCE FROM Client12;
```

Result Grid | Filter Rows: | Export:

BALANCE

DML QUESTIONS:

Part A : Transactions and Savepoints(CLASS Table)

1. COMMIT, SAVEPOINT A, SAVEPOINT B, SAVEPOINT C, SELECT ALL FROM CLASS

CODE:

```
INSERT INTO CLASS VALUES (5,'Rahul');
COMMIT;

UPDATE CLASS SET name = 'Abhijeet' WHERE id='5';
SAVEPOINT A;

INSERT INTO CLASS VALUES (6,'Chris');
SAVEPOINT B;

INSERT INTO CLASS VALUES (7,'Bravo');
SAVEPOINT C;

SELECT*FROM CLASS;
```

OUTPUT:

Result Grid | Filter R

ID	Name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
7	Bravo
NULL	NULL

2. **ROLLBACK TO B , SELECT ALL FROM CLASS**

```
ROLLBACK TO B;
SELECT*FROM CLASS;
```

OUTPUT:

Result Grid

ID	Name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
NULL	NULL

3. ROLLBACK TO A, SELECT ALL FROM CLASS

```
ROLLBACK TO A;
SELECT*FROM CLASS;
```

OUTPUT:

Result Grid

ID	Name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
NULL	NULL

PART B: Retriving Records (Employees table)

1. Find out the employee id, names, and salaries of all the employees.

```
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;
```

OUTPUT:

Result Grid | Filter Rows: | Expo

Employee_Id	First_Name	Last_Name	Salary
101	Akansha	Yadav	500000
102	Dhruv	Kumar	100000
103	Kishan	Verma	900000
104	Neha	Yadav	850000
105	Simon	Jangra	220000

2. List out the employees who under under manager 100.

```
SELECT * FROM EMPLOYEES
WHERE Manager_Id = 100;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	Akansha	Yadav	ay@gmail.com	1234567890	2023-08-16	HR	500000	NULL	100	15
101	Akansha	Yadav	ay@gmail.com	1234567890	2023-08-16	HR	500000	NULL	100	15
103	Kishan	Verma	kv@gmail.com	1234567892	2021-01-09	Law	900000	NULL	100	16
105	Simon	Jangra	simon@gmail.com	1234567894	2023-09-09	Finance	220000	NULL	100	7

3.Find the names of the employees who have a salary greater than or equal to 4800.

CODE:

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;
```

OUTPUT:

Result Grid | Filter Rows:

First_Name	Last_Name
Akansha	Yadav
Dhruv	Kumar
Kishan	Verma
Neha	Yadav
Simon	Jangra

4. List out the employees whose last name is 'AUSTIN'.

CODE and OUTPUT:

```
78 •  SELECT First_Name, Last_Name
79    FROM EMPLOYEES
80    WHERE Last_Name = 'AUSTIN';
```

Result Grid | Filter Rows:

First_Name	Last_Name

5. Find the names of the employees who work in department 60,70 and 80.

CODE and OUTPUT:

```
82 •  SELECT First_Name, Last_Name
83    FROM EMPLOYEES
84    WHERE Department_Id IN (60,70,80);
```

Result Grid | Filter Rows: | Exp

First_Name	Last_Name
Kishan	Verma
Simon	Jangra

6. Display the unique Manager_Id.

CODE and OUTPUT:

```
86 •  SELECT DISTINCT Manager_Id
87    FROM EMPLOYEES;
```

Result Grid | Filter Rows:

Manager_Id
100
77
1
90
19
101
102

PART C: Updating Records (Client_Master Table)

1. Insert five records into Client_Master table.

CODE:

```
INSERT INTO Client_master VALUES ('C123','Rohan','Davarka','Delhi','Delhi',110001,5000);
INSERT INTO Client_master VALUES ('C124','Aman','Park Street','Kolkata','WB',700016,6000);
INSERT INTO Client_master VALUES ('C125','Neha','Link Road','Mumbai','MH',400001,4500);
INSERT INTO Client_master VALUES ('C126','Priya','Sector 15','Noida','UP',201301,7000);
INSERT INTO Client_master VALUES ('C127','Karan','Banjara Hills','Hyderabad','TS',500034,3000);
COMMIT;
SELECT*FROM Client_master;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Con

ClientNo	Name	Address	City	State	Pincode	Bal_due
C123	Rohan	Davarka	Delhi	Delhi	110001	5000
C124	Aman	Park Street	Kolkata	WB	700016	6000
C125	Neha	Link Road	Mumbai	MH	400001	4500
C126	Priya	Sector 15	Noida	UP	201301	7000
C127	Karan	Banjara Hills	Hyderabad	TS	500034	3000

2. Find the names of Clients whose bal_due>5000.

CODE:

```
SELECT Name
FROM Client_master
WHERE Bal_due > 5000;
```

OUTPUT:

Result Grid

Name
Aman
Priya

3. Change the bal_due of Client no “c123” to RS.5100.

CODE and OUTPUT:

```
UPDATE Client12
SET Bal_due = 5100
WHERE ClientNo = 'C123';
```

Result Grid | Filter R

ClientNO	Bal_due
C123	5100
NULL	NULL